

Leann Alhashishi

Machine Learning Researcher

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EDUCATION

University of Cambridge

Cambridge, UK

Master's in Machine Learning and Machine Intelligence

Fall 2025 – Summer 2026

- **Dissertation:** Performing Memory-Intensive Tasks with Test-Time Training
- **Relevant Coursework:** Advanced Machine Learning, Deep Learning, Probabilistic Machine Learning, Advanced Computer Vision, Robotics

University of Maryland, Baltimore County (UMBC)

Baltimore, MD

Bachelor of Computer Science, Magna Cum Laude

Fall 2021 – Spring 2025

RESEARCH & PROJECTS

Performing Memory-Intensive Tasks with Test-Time Training

2026

- **First-author paper in preparation for ICRA 2027**
- Designed a test-time memory pipeline that fits per-episode neural implicit representations (NIRs) via MAML meta-initialization and projects them as tokens directly into a TinyLlama-1.1B, with no frame retrieval required
- Improved video QA accuracy by up to +16.1pp over chance across 96K episodes, and decoded the same NIR weights into continuous robot actions, achieving $R^2 + 0.51$ on 8-dim joints

Upper-Limb Synergy Modeling from Real-World Contactless Motion Tracking

2026

- **IEEE BIOROB 2026 (Accepted, Oral Presentation) - 2nd Author**
- Reduced synergy dimensionality by ~50% (7-10 to 3-5) and captured ~10% more variance than PCA ($p < 1e-7$ across 40 subjects) using Robust PCA on noisy vision-tracked hand motion, validated on a 6-DoF robotic hand

Neural Long-Term Memory for Enterprises

2026

- Building a weight-space implicit memory model as a RAG alternative for enterprise QA
- Using 100x fewer context tokens and 55x faster retrieval vs. dense retrieval over 5K synthetic documents

Co-designing a Social Biosignal Sharing App for Emotional Awareness

2026

- **CHI 2027 (Under Review) - 3rd Author**

Label-Free Novelty Detection and Triage in Smart-Grid Time Series

2025

- Designed a supervised + unsupervised novelty detection pipeline on 1,037 smart-grid users (42K+ samples) using XGBoost/ELSTM+RF + LOF, with clustering-based severity scoring for automated novelty risk triage

WORK EXPERIENCE

Software Engineer Intern

Summer 2024

Cisco

Baltimore, MD

- Implemented production-ready PowerShell MFA modules integrating Duo APIs into enterprise authentication pipelines supporting 2M+ users
- Designed secure Windows SFTP authentication flow for enterprise environments

Engineer Intern

Summer 2023

National Nuclear Security Administration (NNSA)

Washington D.C.

- Designed and automated an end-to-end data pipeline in Python processing 100K+ engineering records, reducing manual analysis time by 30%

SKILLS & CERTIFICATIONS

Frameworks: PyTorch (TorchDynamo, Custom Autograd), JAX, HuggingFace

Computer Vision: 3D Gaussian Splatting, NeRFs, Neural Implicit Representations, COLMAP (SfM), OpenCV

Systems & Infra: Docker, AWS, CI/CD, distributed training (multi-GPU, SLURM)

Languages: Python, C++, SQL, Bash/Shell, CMake